

Bean and rice meals reduce postprandial glycemic response in adults with type 2 diabetes: a cross-over study

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Background Around the world, beans and rice are commonly consumed together as a meal. With type 2 diabetes increasing, the effect of this traditional diet pattern on glycemic response has not been studied fully. **Methods** We evaluated the glycemic response of bean and rice traditional meals compared to rice alone in adults with type 2 diabetes. Seventeen men and women with type 2 diabetes controlled by metformin ($n = 14$) or diet/exercise ($n = 3$) aged 35–70 years participated in the randomized 4×4 crossover trial. The white long grain rice control, pinto beans/rice, black beans/rice, red kidney beans/rice test meals, matched for 50 grams of available carbohydrate, were consumed at breakfast after a 12 hour fast. Capillary blood glucose concentrations at baseline and at 30 minute intervals up to 180 minutes postprandial were collected. MANOVA for repeated measures established glucose differences between treatments. Paired t tests identified differences between bean types and the rice control following a significant MANOVA. **Results** Postprandial net glucose values were significantly lower for the three bean/rice treatments in contrast to the rice control at 90, 120 and 150 minutes. Incremental area under the curve values were significantly lower for the pinto and black bean/rice meals compared to rice alone, but not for kidney beans. **Conclusions** Pinto, dark red kidney and black beans with rice attenuate the glycemic response compared to rice alone. Promotion of traditional foods may provide non-pharmaceutical management of type 2 diabetes and improve dietary adherence with cultural groups. Trial registration Clinical Trials number NCT01241253